

Tutorial 2 Week 4

Tutorial Team 26

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September 4, 2026

Problem 1: Utility Matchmaker

Part A

For any utility function U , the expected utilities of the three lotteries are

$$EU(L_1) = U(1), \quad EU(L_2) = \frac{1}{2}U(2) + \frac{1}{2}U(0), \quad EU(L_3) = U(2).$$

Substituting each of the given utility functions gives

$U(x)$	$EU(L_1)$	$EU(L_2)$	$EU(L_3)$	Relevant preference
x^2	1	2	4	$L_2 \succ L_1$
x	1	1	2	$L_2 \sim L_1$
$\sqrt{x}$	1	$\sqrt{2}/2$	$\sqrt{2}$	$L_1 \succ L_2$

Therefore, the correct matching is

Adam-(b), Bing-(c), Candy-(d), Diana-(a).

Candy requires $L_3 \sim L_2$. This implies

$$U(2) = \frac{1}{2}U(2) + \frac{1}{2}U(0) \iff U(2) = U(0).$$

None of the utility functions in (a)–(c) satisfies this equality, so Candy’s answer is (d), “None of the above.”

Part B

Yes. The two agents have the same preferences. For any lottery L , since the probabilities of all outcomes sum to 1,

$$\begin{aligned} EU_1(L) &= \mathbb{E}_L[U_1] = \mathbb{E}_L[k_1U_2 + k_2] \\ &= k_1\mathbb{E}_L[U_2] + k_2 = k_1EU_2(L) + k_2. \end{aligned}$$

Therefore, for any lotteries L and M , since $k_1 > 0$,

$$EU_1(L) \geq EU_1(M) \iff EU_2(L) \geq EU_2(M).$$

A positive affine transformation does not change the ranking of lotteries, including strict preferences and indifference relations.

Problem 2: Adventures of PacBaby

The probability of theft is 0.2, the probability of no theft is 0.8, and PacBaby's utility function is $U(x) = \ln(1 + x)$.

(a) Without Insurance

If no theft occurs, PacBaby's wealth is 100. If theft occurs, her wealth is 0. Therefore,

$$\begin{aligned} EU(\text{no insurance}) &= 0.8U(100) + 0.2U(0) \\ &= 0.8 \ln(101) + 0.2 \ln(1) \\ &= \boxed{0.8 \ln(101) \approx 3.6921}. \end{aligned}$$

(b) With Insurance

After paying the premium, PacBaby's final wealth is 70 regardless of whether theft occurs. Hence,

$$\begin{aligned} EU(\text{insurance}) &= 0.8U(70) + 0.2U(70) \\ &= \boxed{\ln(71) \approx 4.2627}. \end{aligned}$$

Since

$$\ln(71) \approx 4.2627 > 0.8 \ln(101) \approx 3.6921,$$

PacBaby should buy the insurance in order to maximize her expected utility.

(c) Expected Monetary Value for PacIncome

PacIncome always receives the \$30 premium and pays \$70 with probability 0.2. Its expected net monetary gain from selling one policy is therefore

$$EMV = 30 - 0.2(70) = 30 - 14 = \boxed{\$16}.$$

Problem 3: Allais Paradox

Let

$$u_4 = U(4000), \quad u_3 = U(3000), \quad U(0) = 0.$$

The expected utilities of the four lotteries are

$$EU(A) = 0.8u_4, \quad EU(B) = u_3, \quad EU(C) = 0.2u_4, \quad EU(D) = 0.25u_3.$$

(a) Contradiction Under Expected-Utility Analysis

The preference $B \succ A$ requires

$$u_3 > 0.8u_4. \tag{1}$$

On the other hand, the preference $C \succ D$ requires

$$0.2u_4 > 0.25u_3.$$

Multiplying both sides by 4 gives

$$0.8u_4 > u_3, \tag{2}$$

which is the exact opposite of (1). Therefore, no expected-utility function can represent both $B \succ A$ and $C \succ D$. A rational agent who follows expected-utility theory cannot make both choices simultaneously.

More specifically, if $B \succ A$, then it must be the case that $D \succ C$. Conversely, if $C \succ D$, then it must be the case that $A \succ B$.

(b) Violation of the Substitutability Axiom

Let Z denote the lottery that pays \$0 with certainty, and define

$$M_A = [0.25, A; 0.75, Z], \quad M_B = [0.25, B; 0.75, Z].$$

By decomposability, M_A pays \$4000 with probability $0.25 \times 0.8 = 0.20$ and \$0 otherwise, while M_B pays \$3000 with probability 0.25 and \$0 otherwise. Hence,

$$M_A \sim C, \quad M_B \sim D.$$

Starting from the observed preference $B \succ A$, substitutability requires this preference to be preserved when both lotteries are mixed with the same lottery Z using the same probabilities. Thus,

$$B \succ A \implies M_B \succ M_A \implies D \succ C.$$

The last implication follows from decomposability and transitivity. It contradicts the observed preference $C \succ D$. Therefore, the judgments $B \succ A$ and $C \succ D$ jointly violate substitutability.

For comparison, a cancellation proof would start from $C \succ D$ and remove the common 0.75 probability of Z to infer $A \succ B$, contradicting $B \succ A$. Our proof instead introduces the common branch, directly applying the substitutability axiom requested in the question.

Generative AI Acknowledgement

OpenAI Codex (GPT-5) was used as a discussion and refinement aid while preparing this document. Its main uses were to provide high-level guidance and hints during discussion, and to refine the wording and $\text{\LaTeX}$ formatting. The authors reviewed and verified the final solutions and take responsibility for their correctness. No search engine or other generative AI model was used in preparing this document.